
AAD-50

USER MANUAL

Windows GUI Application

PRODUCT : The Abeselom ASIC-Direct 50 (AAD-50)

VERSION : 1.0.2

PLATFORM : Windows 10 1607+ / Windows 11

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REPOSITORY : github.com/yonasabeselom/aad50

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LICENCE : Proprietary -- Personal use only

This document provides complete operating instructions for the AAD-50 Windows GUI application. Read Section 11 (Safety) before performing any live data destruction operation.

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SECTION 01 == INTRODUCTION =====

AAD-50 is an open-source, firmware-enforced NVMe solid-state drive sanitization framework. It bypasses the operating system entirely and communicates directly with the drive controller ASIC via raw IOCTL commands, reaching physical NAND cells invisible to standard tools.

WHY STANDARD TOOLS FAIL ON SSDs

```
+-----+
| Problem: DoD 5220.22-M / Gutmann 35-pass |
| ----- |
| Designed for magnetic HDDs. Intercepted by the |
| Flash Translation Layer (FTL) on NVMe SSDs. |
| |
| 67% of SSD data survives standard overwrite. |
| (University of California San Diego, 2011) |
| |
| AAD-50 bypasses the FTL entirely. |
+-----+
```

DESTRUCTION MATRIX

PHASE	CYCLES	ACTION	CDW10
B	01-40	Physical NAND Cell Overwrite	0x02
C	41-45	Flash Translation Layer Reset	0x01
A	46-50	Cryptographic Key Destruction	0x04

Each cycle is hardware-confirmed via Log Page 0x81 (SSTAT) polling.

SECTION 02 == SYSTEM REQUIREMENTS =====

REQUIREMENT	VALUE	NOTES
Operating System	Windows 10 v1607+	Win 11 supported
Privileges	Administrator	Required
Drive Type	NVMe SSD	SATA not supported
Python (optional)	3.10+	Not needed for .exe
Disk Space	50 MB	App + reports

SECTION 03 == INSTALLATION =====

OPTION A -- STANDALONE EXE (RECOMMENDED)

No Python required. Runs on any Windows 10/11 machine.

- [1] Go to: github.com/yonasabeslom/aad50/releases
- [2] Download AAD50.exe from the latest release
- [3] Right-click AAD50.exe --> Run as administrator

OPTION B -- PYTHON SCRIPT

```
C:\> pip install customtkinter reportlab
C:\> cd C:\path\to\AAD50
C:\AAD50> python aad50_gui_windows.py
```

NOTE: aad50_gui_windows.py and aad50_abeslom_windows.py
must be in the same folder.

SECTION 04 === GETTING STARTED -- FIRST RUN ==

Complete walkthrough for your first simulation run.

Dry-Run mode is ON by default. Your drive is safe.

- | | |
|------|--|
| [01] | Launch Right-click AAD50.exe --> Run as administrator |
| [02] | Verify safe mode Header must show: SIMULATION ACTIVE (amber) |
| [03] | Select Drive Click 'Select Drive' in the left sidebar |
| [04] | Choose target Click 'Select Drive' next to your NVMe drive |
| [05] | Continue Click 'Continue to Sanitize' button |
| [06] | Operator Name Type your name in the Operator Name field |
| [07] | Authorize Type exactly: EXECUTE-AAD-50-ABESELOM |
| [08] | Execute Click 'RUN SIMULATION (DRY-RUN -- SAFE)' |
| [09] | Results View completion screen with SHA-256 audit hash |
| [10] | Save report Click 'PDF Certificate' to save your certificate |

SECTION 05 == THE FIVE SCREENS =====

■ HOME DASHBOARD

Main overview screen.

Shows: phase matrix, 50 cycles, 3 phases, opcode 0x84, SHA-256.

Click 'Start -- Select Your Drive' to begin.

■ SELECT DRIVE

Detects all NVMe drives automatically.

Shows: drive model, device path, volume letters.

First scan: 1-2 seconds. Subsequent: instant (cached).

Click 'Select Drive' then 'Continue to Sanitize'.

■ SANITIZE DRIVE

Main operational screen.

Left: warning description (simulation=amber / live=red).

Right: drive info, operator name, auth token, execute.

Dry-Run toggle in sidebar controls mode.

■ AUDIT REPORTS

Load and verify saved JSON audit reports.

Recalculates SHA-256 hash from cycle records.

VERIFIED = report is genuine and unaltered.

FAILED = report was modified after generation.

■ ABOUT

Tool info, version, author, compliance standards.

Licence information and contact details.

OEM Driver Diagnostic: tests Log Page 0x81

pass-through capability of your NVMe driver.

SECTION 06 == DRY-RUN SIMULATION MODE =====

Test AAD-50 safely. Zero hardware commands sent.

WHAT SIMULATION MODE DOES:

- [+] Runs all 50 cycles -- identical timing and logging to live
- [+] Skips all DeviceIoControl IOCTL calls completely
- [+] Generates real SHA-256 audit hash from simulated records
- [+] Never opens drive handle -- OS, files, drive untouched
- [+] Produces full JSON report and PDF Certificate

MODE INDICATORS:

LOCATION	SIMULATION MODE	LIVE MODE
Header badge	SIMULATION ACTIVE	REAL WIPE ARMED
Badge color	amber	red
Sidebar label	Simulation active -- safe mode	LIVE MODE -- REAL DATA WILL BE DESTROYED
Execute button	RUN SIMULATION (DRY-RUN -- SAFE)	EXECUTE AAD-50 SANITIZATION -- LIVE

SECTION 07 === LIVE SANITIZATION -- REAL HARDWARE

```
+-----+
| /!\ WARNING |
| |
| LIVE MODE PERMANENTLY AND IRREVERSIBLY DESTROYS ALL DATA. |
| NO RECOVERY IS POSSIBLE. |
| ONLY PROCEED IF YOU ARE ABSOLUTELY CERTAIN. |
+-----+
```

PRE-FLIGHT CHECKLIST:

- [] Target drive contains NO data you want to keep
- [] Target drive is NOT your Windows boot drive (C:)
- [] Running as Administrator
- [] Operator Name ready to enter
- [] Save location ready for audit report

EXECUTION STEPS:

- [01] Toggle Dry-Run OFF in sidebar (switch turns grey)
- [02] Verify: header shows REAL WIPE ARMED [DESTRUCTIVE]
- [03] Click Select Drive
- [04] Confirm correct drive -- check model AND serial number
- [05] Click Continue to Sanitize
- [06] Read the full IRREVERSIBLE DATA DESTRUCTION WARNING
- [07] Enter operator name
- [08] Type: EXECUTE-AAD-50-ABESEL0M
- [09] Click EXECUTE AAD-50 SANITIZATION -- LIVE
- [10] Confirm second popup -- click Yes
- [11] Monitor 50-cycle progress (phases B, C, A)
- [12] Save PDF Certificate on completion screen

SECTION 08 == AUDIT REPORTS =====

Every run produces a JSON audit report sealed with SHA-256.

JSON REPORT FIELDS:

FIELD	DESCRIPTION
<i>tool</i>	Tool name and version
<i>device</i>	Drive path e.g. \\.\PhysicalDrive0
<i>drive_model</i>	Manufacturer and model
<i>drive_serial</i>	Drive serial number
<i>operator</i>	Operator name entered at runtime
<i>started_at</i>	UTC start timestamp
<i>completed_at</i>	UTC completion timestamp
<i>dry_run</i>	true (simulation) or false (live)
<i>cycles</i>	Array of 50 cycle records
<i>outcome</i>	SUCCESS -- DATA DESTROYED or FAILED
<i>log_hash</i>	SHA-256 hash of all cycle records

VERIFYING A REPORT:

- [1] Click 'Audit Reports' in the sidebar
- [2] Click 'Load JSON Report'
- [3] Select your .json file
- [4] Check result: VERIFIED = genuine | FAILED = tampered

SECTION 09 == PDF CERTIFICATE OF DESTRUCTION

Professional compliance document for GDPR, HIPAA, ISO 27001 audits.

REQUIREMENT:

```
C:\> pip install reportlab
```

CERTIFICATE CONTENTS:

- [+] Outcome banner (green=SUCCESS / red=FAILED)
- [+] Drive model, serial number, device path, volume letters
- [+] All 3 phases with cycle counts and timestamps
- [+] Operator name and execution mode
- [+] NIST SP 800-88 Rev.2 / IEEE 2883-2022 / ISO/IEC 27040
- [+] SHA-256 audit hash (bright green on dark background)
- [+] Author, GitHub link, validity statement

HOW TO GENERATE:

- [1] Complete a sanitization run (simulation or live)
- [2] On completion screen: click 'PDF Certificate'
- [3] Save dialog defaults to Documents folder
- [4] Open PDF and verify serial number, operator, hash

SECTION 10 == TROUBLESHOOTING =====

```
+-----+
| PROBLEM : No drives appear |
| CAUSE : Not running as Administrator. |
| FIX : Right-click --> Run as administrator |
+-----+
```

```
+-----+
| PROBLEM : NameError on launch |
| CAUSE : Outdated .py file. |
| FIX : Download latest from github.com/yonasabesalom/aad50 |
+-----+
```

```
+-----+
| PROBLEM : PDF generation fails |
| CAUSE : ReportLab not installed. |
| FIX : pip install reportlab |
+-----+
```

```
+-----+
| PROBLEM : PDF permission denied |
| CAUSE : Saving to restricted location. |
| FIX : Save to Documents folder |
+-----+
```

```
+-----+
| PROBLEM : Drive scan slow (1-2s) |
| CAUSE : Normal on first visit -- cached after. |
| FIX : Expected. Only refresh when drives change. |
+-----+
```

```
+-----+
| PROBLEM : Auth token rejected |
| CAUSE : Wrong capitalisation or hyphens. |
| FIX : Type: EXECUTE-AAD-50-ABESELOM exactly |
+-----+
```

```
+-----+
| PROBLEM : Driver test fails |
| CAUSE : NVMe driver lacks Log Page 0x81 support. |
| FIX : Update NVMe driver from manufacturer website |
+-----+
```

SECTION 11 == SAFETY AND WARNINGS =====

```
+-----+
| /!\ WARNING |
| |
| READ THIS SECTION BEFORE PERFORMING ANY |
| LIVE SANITIZATION. DATA LOSS IS PERMANENT. |
+-----+
```

ABSOLUTE RULES:

- [!] NEVER run live mode on a drive with data you want to keep
- [!] NEVER run live mode on your Windows boot drive (C:)
- [!] ALWAYS verify drive model AND serial number before executing
- [!] ALWAYS run dry-run simulation first on any new machine
- [!] ALWAYS save audit report and PDF certificate after execution
- [!] ALWAYS run as Administrator

COMPLIANCE STANDARDS:

STANDARD	RELEVANCE
NIST SP 800-88 Rev.2	Purge class for solid-state media
IEEE 2883-2022	Storage device sanitization
ISO/IEC 27040:2015	Storage security guidelines
GDPR Article 17	Right to erasure
HIPAA	Medical data disposal requirements